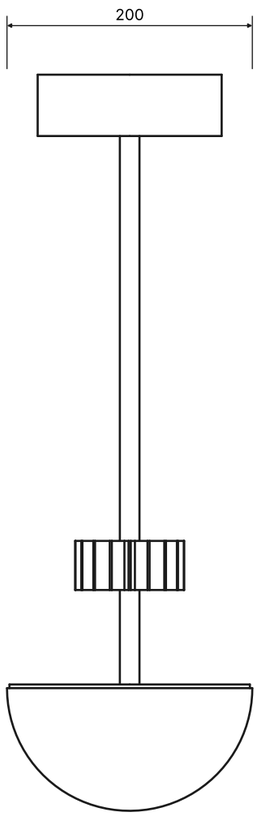
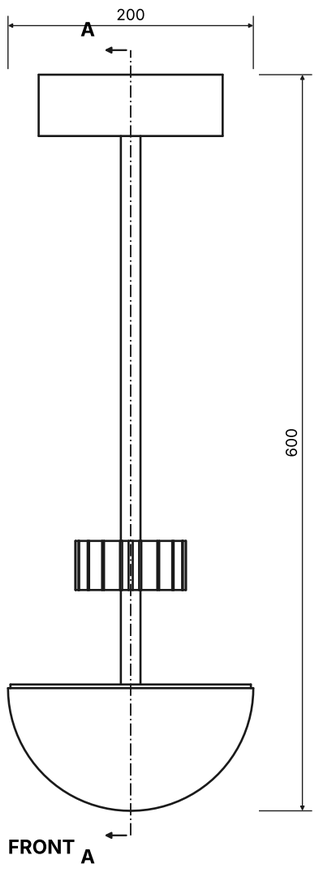
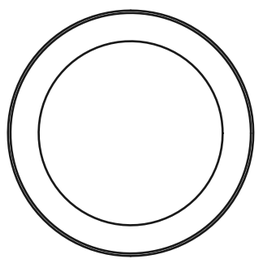
Scale 1:5. First angle: the view from the right is drawn to the left of the front, the view from above under it. A-A: the section on sheet 2.



FROM THE RIGHT



FRONT



FROM ABOVE

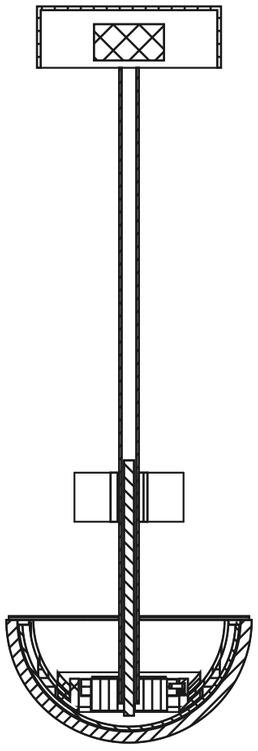
General arrangement

Hemisphere South: Uplighter bowl on a rod in cast acrylic, its own shade, nothing below it

mm | issue 2026-10-23 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. Section A-A (sheet 1): the product cut at x 0.0 and seen from the right; cut faces hatched, a different angle for each part.



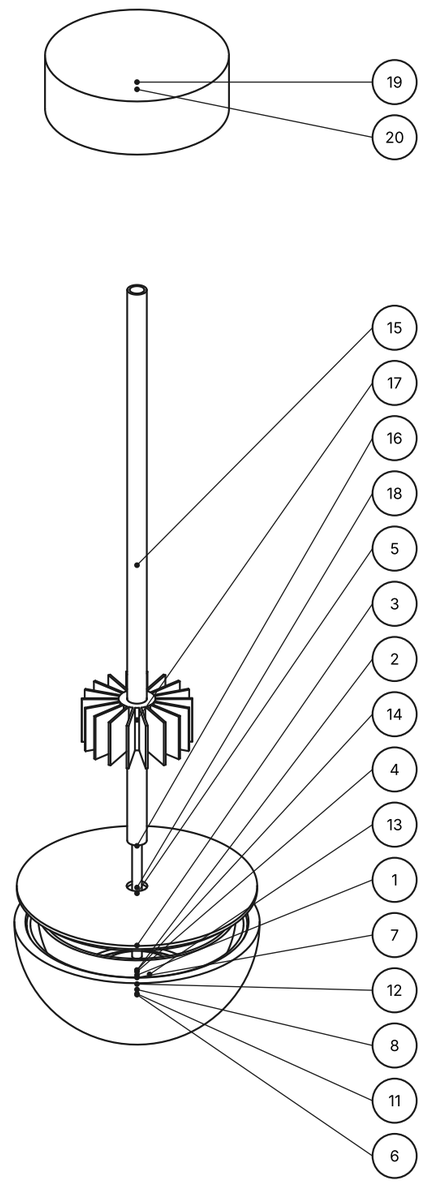
Section A-A

Hemisphere South: Uplighter bowl on a rod in cast acrylic, its own shade, nothing below it

mm | issue 2026-10-23 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5, parts drawn apart. Balloons are the parts list's numbers (last sheet).



Exploded view

Hemisphere South: Uplighter bowl on a rod in cast acrylic, its own shade, nothing below it

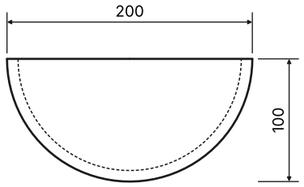
mm | issue 2026-10-23 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

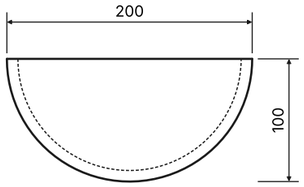
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

1. shell: formed, acrylic-formed; 516 cm3 each; 1 for the set; 1:5

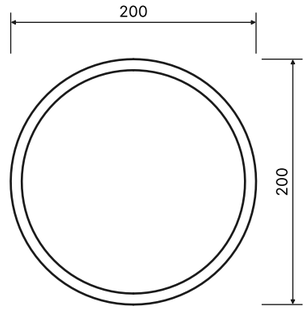
South's own: the outside, closed at the crown (North's shell mould, untrimmed there): 10 mm tinted cast sheet drape-formed from cast sheet over an aluminium mould, trimmed and polished; the prototypes are formed on the production moulds



FRONT



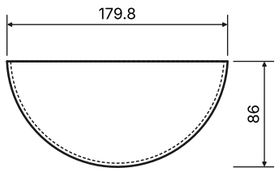
FROM THE RIGHT



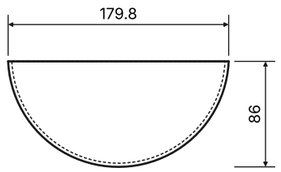
FROM ABOVE

2. liner: formed, acrylic-opal-formed; 141 cm3 each; 1 for the set; 1:5

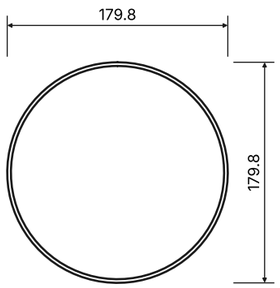
3 mm opal, closed at the crown (North's liner mould): it hides the collar's flange from below; drape-formed from cast sheet over an aluminium mould, trimmed and polished; the prototypes are formed on the production moulds



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Hemisphere South: Uplighter bowl on a rod in cast acrylic, its own shade, nothing below it

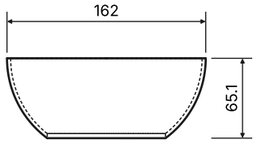
mm | issue 2026-10-23 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

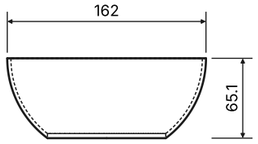
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

3. skin: formed, acrylic-formed; 95 cm3 each; 1 for the set; 1:5

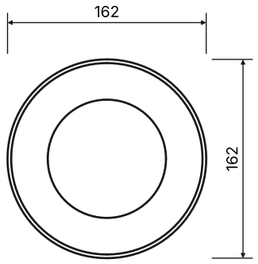
South's own mould (6 in from the liner): 3 mm tinted cast sheet, the crown hole for the collar; it carries the shade on the collar's flange and clamp ring



FRONT



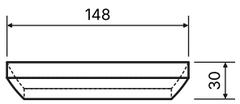
FROM THE RIGHT



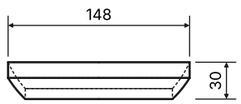
FROM ABOVE

4. weld: lathe, cast-acrylic; 60 cm3 each; 1 for the set; 1:5

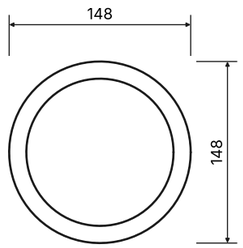
a tinted acrylic ring turned to fill the gap round the flange: the skin, this ring and the liner welded with polymerising acrylic cement (Acrifix 192 kind); the shell and the liner welded at the rim



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Hemisphere South: Uplighter bowl on a rod in cast acrylic, its own shade, nothing below it

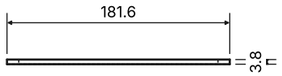
mm | issue 2026-10-23 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

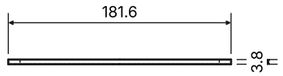
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

5. rim: lathe, cast-acrylic; 20 cm3 each; 1 for the set; 1:5

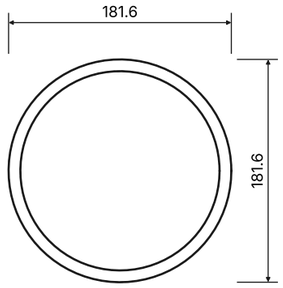
closes the three layers at the mouth, welded and polished with them; a press-in ePTFE vent in its face



FRONT



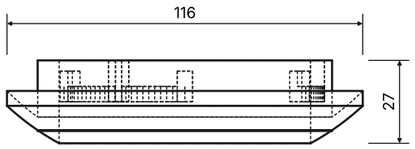
FROM THE RIGHT



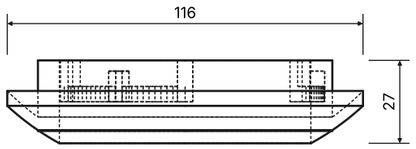
FROM ABOVE

6. collar: lathe, aluminium; 61 cm3 each; 1 for the set; 1:2

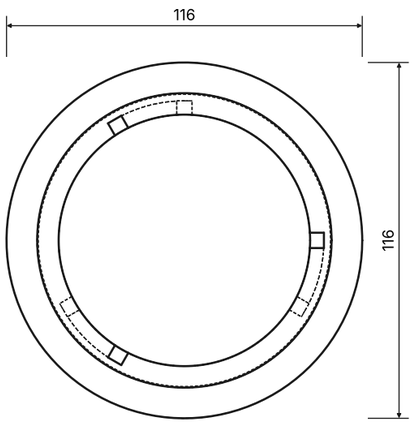
aluminium 6061, black anodised: its flange on the skin's outside, behind the opal (unseen); its tube inside the bowl, with three L-slots open at the inner end for South's pins



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Hemisphere South: Uplighter bowl on a rod in cast acrylic, its own shade, nothing below it

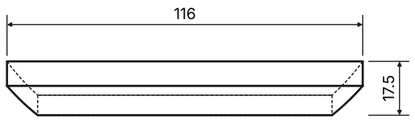
mm | issue 2026-10-23 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

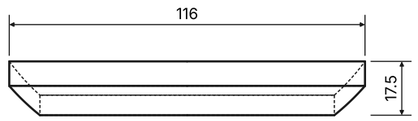
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

7. clamp: lathe, aluminium; 24 cm3 each; 1 for the set; 1:2

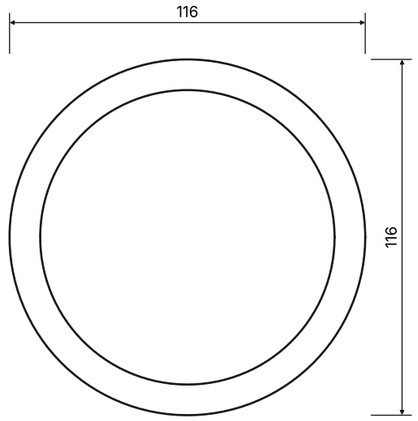
aluminium, black anodised: screwed onto the collar inside the bowl, on silicone, clamping the skin



FRONT



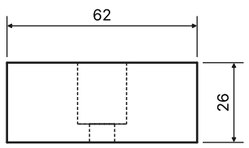
FROM THE RIGHT



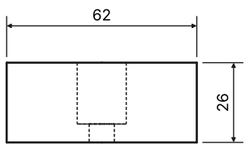
FROM ABOVE

11. hub: lathe, aluminium; 74 cm3 each; 1 for the set; 1:2

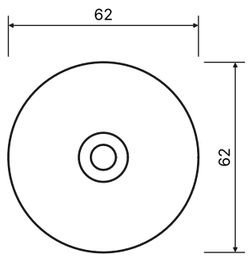
aluminium 6061, black anodised: the core, turned from bar, inside the collar; its outer face behind the opal, unseen; the LED ring on its inner face; the rod in a 20 deep socket, the heat pipe's evaporator in its bore on thermal paste



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Hemisphere South: Uplighter bowl on a rod in cast acrylic, its own shade, nothing below it

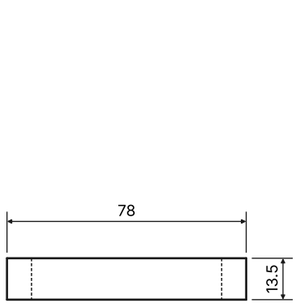
mm | issue 2026-10-23 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

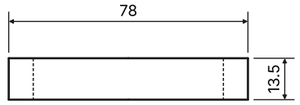
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

12. ring: lathe, peek; 24 cm3 each; 1 for the set; 1:2

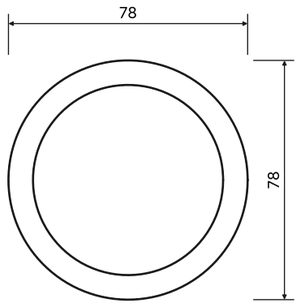
PEEK, unfilled: the thermal break, at the collar's inner end; it carries the pins (into the collar's second set of slots) and the detent



FRONT



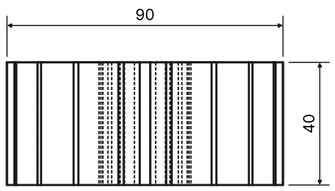
FROM THE RIGHT



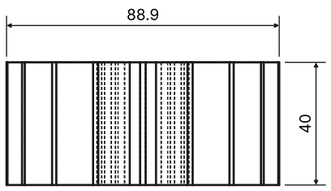
FROM ABOVE

17. fins: extrusion, aluminium; 59 cm3 each; 1 for the set; 1:2

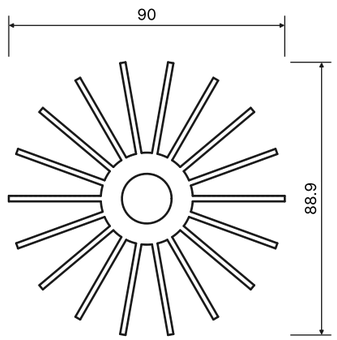
the heat sink, on the rod 80 above the lid, in room air: 40 cut from a radial extrusion 90 across, 18 fins, black anodised, clamped on the rod over the heat pipe's condenser; its warm air rises to the ceiling, away from the acrylic



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Hemisphere South: Uplighter bowl on a rod in cast acrylic, its own shade, nothing below it

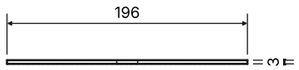
mm | issue 2026-10-23 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

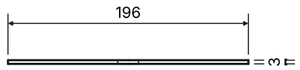
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

18. lid: mill, cast-acrylic; 90 cm3 each; 1 for the set; 1:5

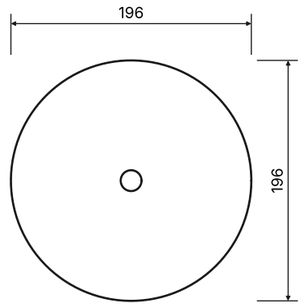
3 mm clear cast acrylic, bead-blasted both faces: the dust lid, lying on the rim on three small clear silicone bumpers; it lifts up the rod to be wiped. Its light is the ceiling wash



FRONT



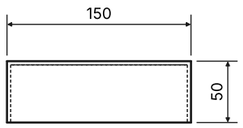
FROM THE RIGHT



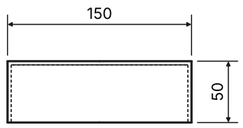
FROM ABOVE

19. canopy: lathe, cast-acrylic; 118 cm3 each; 1 for the set; 1:5

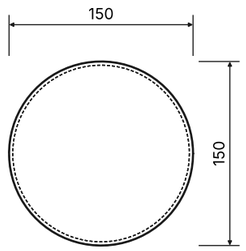
150 × 50, black cast acrylic, polished, as North's: over the driver and a steel junction-box strap; the rod hangs from the strap through a listed swivel so it hangs plumb



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Hemisphere South: Uplighter bowl on a rod in cast acrylic, its own shade, nothing below it

mm | issue 2026-10-23 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

For 1; prices are budgets (bom.md says where each comes from).

no.	item	make	qty	what	size	each USD
1	shell	formed	1	Cast acrylic sheet, tinted through, drape-formed over an aluminium mould	516 cm3, 210 × 210 × 110	1,539.06
2	liner	formed	1	Cast acrylic sheet, volume-diffusing (opal), formed over an aluminium mould	141 cm3, 190 × 190 × 96	1,214.30
3	skin	formed	1	Cast acrylic sheet, tinted through, drape-formed over an aluminium mould	95 cm3, 172 × 172 × 75	1,211.25
4	weld	lathe	1	Cast acrylic tube, tinted through	158 × 158 × 40	84.96
5	rim	lathe	1	Cast acrylic tube, tinted through	192 × 192 × 14	65.26
6	collar	lathe	1	Aluminium 6061	126 × 126 × 37	59.70
7	clamp	lathe	1	Aluminium 6061	126 × 126 × 28	55.93
8	gaskets	buy	1	Two moulded silicone rings (Shore 40, black, 1 mm, spherical) for South's own shade: one between the collar's flange and the skin's outside, one between the clamp ring and the skin's inside		3.00
9	vent	buy	1	Gore or alike A press-in ePTFE pressure-equalising vent (about 4 across), in the rim ring: it lets the air gaps between the layers breathe and keeps out dust and liquid		1.50
10	welding	buy	1	the shade maker Welding South's shade: the skin, the crown weld ring and the opal liner, and the shell and the liner at the rim, with a polymerising acrylic cement (Acrifix 192 kind), cured, annealed and polished; a pull test of the first shades		8.00
11	hub	lathe	1	Aluminium 6061	72 × 72 × 36	49.67
12	ring	lathe	1	PEEK rod, unfilled	88 × 88 × 24	108.69
13	pins	buy	1	Three stainless dowel pins 4 × 8 (ISO 8734), pressed into the hub: the bayonet's pins		1.50
14	led	buy	1	to make (an LED board house), with Samsung LM301H-class 3030 LEDs Hemisphere South's light: a ring MCPCB 74 × 14 × 1.6 (aluminium core), 2700 K CRI 90 mid-power LEDs in strings matched to the driver's window: about 2,200 lm at about 14.7		14.00
15	rod	buy	1	to make (an anodiser), or a lighting supplier's tube Aluminium tube 16 × 2 (6063), black anodised, cut to the drop, threaded into the hub; the heat pipe and the 18 AWG Class 2 wires inside it; a listed swivel (hickey) and a locknut at the canopy		9.00
16	pipe	buy	1	any (Wakefield, Advanced Thermal Solutions, Shenzhen makers) A sintered copper-water heat pipe, 8 across, about 220 long: its evaporator in the hub (at the bottom: gravity returns the fluid, the easiest way round for a heat pipe), its		6.00
17	fins	extrusion	1	Aluminium 6061	100 × 99 × 50	17.37
18	lid	mill	1	Cast acrylic tube, tinted through	206 × 206 × 13	67.07
19	canopy	lathe	1	Cast acrylic tube, tinted through	160 × 160 × 60	106.44
20	driver	buy	1	Mean Well (to confirm against the board) A Class 2 constant-current, phase-cut dimmable driver, 120 V in, for the ring board: PCD-25 class for the 2,200 lm level (about 15 W), PCD-16 class for the 1,150 lm level (about 8 W); the board's strings		24.00
21	plunger	buy	1	Kipp or Misumi A stainless ball spring plunger M5, light force (about 10 N), threaded into the hub: the bayonet's detent, clicking into a dimple in the collar		4.50
22	listing	buy	1	UL or Intertek (ETL) Listing the pendant to UL 1598 (and the table lamp's battery and charger to UL 153 / IEC 62368): a one-off cost a design, spread over the first run		[PRICE]

Parts list

Hemisphere South: Uplighter bowl on a rod in cast acrylic, its own shade, nothing below it

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Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Notes

1. A bowl, opening up, on a slim rod from the ceiling, with nothing below it (the owner, rounds 36 and 37): its own shade, closed underneath. Outside a tinted shell and an opal liner, both closed; inside them a tinted skin that carries the shade on the collar, whose flange the opal hides. North's and South's shades no longer swap (the owner accepts it).
2. The light: a ring of mid-power LEDs on the hub's inner face at the bottom of the bowl, facing up. About 57 % of it leaves through the mouth and the frosted lid as a soft wash on the ceiling; the rest falls on the bowl's wall, which glows in its tint. Twice North's light: about 1,150 lm.
3. The heat: up a copper heat pipe in the rod to black fins on the rod 80 above the lid, in room air, so the warm air rises to the ceiling, not past the acrylic. The PEEK ring keeps the hub off the collar. See hemisphere_thermal.py and the test plan (README, rounds 35 and 36).
4. The dust lid: 3 mm frosted acrylic on the rim, lifted up the rod to wipe.
5. Hang with the rim 450 to 600 below a ceiling of 2.6 to 3.2 m, so the wash spreads; not over a table (its light is on the ceiling).

Notes

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